RESEARCH PAPER

240,000 plastic particles in 1 litre of bottled water.

You can't see them. They are there.

SWIPE →

WHO TESTED IT?

Columbia University, 2024

Published in PNAS.

Bottled water from 3 popular US brands was tested.

A new laser microscope (SRS) identifies each particle one by one, and tells which plastic it is.

HOW MANY?

110,000 to 370,000

Per litre, about 240,000 on average.

That is 10–100 times more than earlier estimates.

A 2018 study found 325 per litre on average. The difference is simple: we couldn't see the smallest ones before.

WHAT IS A NANOPLASTIC?

About 90% of the particles

Microplastic: from 5 mm down to 1 micrometre.

Nanoplastic: smaller than 1 micrometre. A human hair is roughly 50–100 times thicker.

So small that scientists are studying whether they pass through the gut and lungs into the blood.

WHERE DO THEY COME FROM?

The filter and the bottle itself

The most common was polyamide (a type of nylon), likely from plastic filters used to purify the water.

PET comes from the bottle: squeezing it, heat, opening and closing the cap.

So the thing cleaning the water leaves a trace too.

STILL UNKNOWN

What the other 90% is

The team looked for 7 common plastics. They made up only ~10% of the nanoparticles found.

What the rest is, and how it affects health, is still being studied.

This isn't a reason to panic. It means we can finally count them.

Found this useful? Save it.

You'll need it one day.

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